CLASS 9 · SCIENCE | CHAPTER 5 | TOPIC 2A

Date: ___ / ___ / _____

Name: _____

Cell Structure

Membrane · Wall · Cytoplasm · Nucleus

📖 *The Fundamental Unit of Life — Part 2a*

✅ Cell Membrane | ✅ Cell Wall | ✅ Cytoplasm | ✅ Nucleus | ⏭ Organelles → Topic 2b

🧩 **The Big Idea:** Every cell has a boundary that controls what enters and exits, a jelly-like filling where life happens, and a control centre that holds the genetic instructions. These four structures — membrane, wall, cytoplasm, and nucleus — are the *core architecture* of a cell.

CELL MEMBRANE (PLASMA MEMBRANE)

1 Cell Membrane

Also called: Plasma Membrane · Present in ALL cells

The **thin, flexible outer boundary** of the cell that separates the cell's internal environment from its surroundings. Made of **lipids and proteins**.

- → Called a **Selectively Permeable Membrane** — allows only certain substances to pass in or out.
- → Controls the movement of substances like water, glucose, O_2, and CO_2.
- → Movement of water across it is called **Osmosis**.
- → Movement of other molecules is called **Diffusion**.
- → In animal cells, it is the **outermost boundary**. In plant cells, it lies just inside the cell wall.
- → Flexible nature allows cells to engulf food — process called **Endocytosis** (e.g. Amoeba).

CELL WALL

2 Cell Wall

Present in: Plant cells, Fungi, Bacteria · ABSENT in Animal cells

A **rigid, non-living layer** outside the cell membrane found in plant cells. Made of **Cellulose** (a complex carbohydrate).

- → Provides **structural support and strength** to the plant cell.
- → Protects the cell from mechanical damage and infection.
- → It is **fully permeable** — allows all substances to pass through freely.
- → Gives a **definite shape** to the plant cell.
- → Responsible for the **turgidity** of the cell (firmness due to water pressure).

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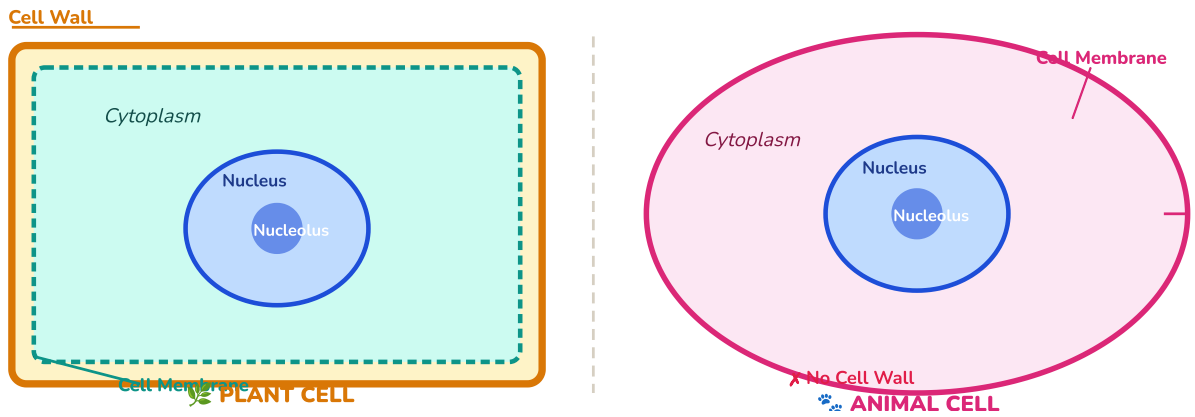
→ In bacteria, made of **peptidoglycan**; in fungi, made of **chitin** — not cellulose.

🔬 DIAGRAM: PLANT CELL VS ANIMAL CELL (BOUNDARY LAYERS)



☝ *Key difference: Plant cells have Cell Wall + Cell Membrane. Animal cells only have Cell Membrane.*

🫗 CYTOPLASM

3 Cytoplasm

Also called: Cytosol (liquid part) · Present in ALL cells

> The **jelly-like, semi-fluid substance** that fills the cell between the nucleus and the cell membrane. It is the *site of most cellular activities*.

→ Made up of water, salts, and various organic molecules.

→ All **cell organelles** (mitochondria, ribosomes, ER, etc.) are embedded in the cytoplasm.

→ Site of many metabolic reactions including **glycolysis** (breakdown of glucose).

→ The fluid part (without organelles) is called **Cytosol**.

→ Contains a network of fibres called the **Cytoskeleton** that gives shape to the cell.

→ Cytoplasm + Nucleus together = **Protoplasm** (the "living substance").

🎯 NUCLEUS

4 Nucleus

The Control Centre of the Cell · Discovered by Robert Brown (1831)

> A **large, spherical organelle** surrounded by a double-layered membrane. It contains the cell's genetic material (DNA) and controls all cellular activities.

→ Surrounded by a **Nuclear Membrane (Nuclear Envelope)** — a double membrane with small pores called **Nuclear Pores**.

→ Contains **Chromatin** — a network of DNA and protein threads. During cell division, chromatin condenses into **Chromosomes**.

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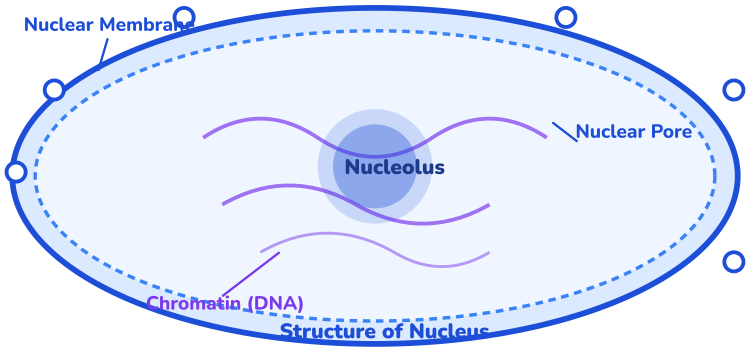
→ Contains a dense, spherical body called the **Nucleolus** — site of ribosome production.

→ **Chromosomes carry genes** — units of inheritance that pass traits from parent to offspring.

→ Cells without a defined nucleus = **Prokaryotes** (e.g. bacteria). Cells with a nucleus = **Eukaryotes**.

→ Mature RBCs (Red Blood Cells) in mammals **have no nucleus** — exception.



☝️ *Label all 4 parts: Nuclear Membrane, Nuclear Pore, Nucleolus, Chromatin*

⚖️ PLANT CELL VS ANIMAL CELL — KEY DIFFERENCES

FEATURE	🌿 PLANT CELL	🐾 ANIMAL CELL
Cell Wall	Present (Cellulose)	Absent
Cell Membrane	Present	Present
Shape	Fixed, rectangular	Irregular, flexible
Vacuole	Large, central vacuole	Small or absent
Plastids	Present	Absent
Centrioles	Absent	Present
Nucleus	Present (peripheral)	Present (central)

⭐ EXAM TIPS

- "Selectively permeable" = most common phrase to write for plasma membrane.
- Osmosis = water movement; Diffusion = other molecules. Don't mix up.
- Prokaryote vs Eukaryote — likely 2–3 marks. Learn definition + examples.
- RBC has no nucleus — very common 1-mark trap question.

⚠️ COMMON MISTAKES

- ❌ Cell wall = selectively permeable → ✅ Cell MEMBRANE is selectively permeable. Cell wall is fully permeable.
- ❌ Cytoplasm = only the liquid → ✅ Cytoplasm includes all organelles too.
- ❌ Nucleus = only in plants → ✅ Nucleus is in ALL eukaryotic cells.
- ❌ Chromatin = same as chromosome → ✅ Chromatin becomes chromosome during cell

- Nuclear membrane = double membrane (not single). Always mention "double."

division.

🧠 **MEMORY TRICKS**

"Cell Wall = PLANT's Wall" — Walls are for houses (plants stay in one place), animals move freely — no wall needed!

Nucleus = "Principal's Office" — Controls everything that happens in school (the cell). Has a secretary (Nucleolus) who makes the ribosomes (workers).

Protoplasm = Proto + Plasma → "First living substance" (proto = first, plasma = living matter)

🏷️ KEYWORDS

Plasma Membrane | Selectively Permeable | Osmosis | Diffusion | Endocytosis | Cell Wall | Cellulose | Turgidity | Cytoplasm | Cytosol | Protoplasm | Nucleus | Nuclear Pore | Chromatin | Nucleolus | Prokaryote | Eukaryote

✍️ SELF-TEST

1. What is meant by "selectively permeable"? Give one example. [_____]
2. Name the material that makes up the plant cell wall. [_____]
3. What is the difference between cytoplasm and cytosol? [_____]
4. Who discovered the nucleus? What does it contain? [_____]
5. Name ONE animal cell that has no nucleus. Why? [_____]
6. What is the difference between chromatin and chromosomes? [_____]

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